## Experiment 15: EasyToll – Online Toll Payment System

**Aim:**

To develop an online toll payment website that lets users manage toll accounts, check balances, purchase toll e-tickets, apply for toll passes, and generate QR codes.

**Algorithm:**

1. Display toll account details and available toll services.
2. Allow users to enter vehicle details and add money to an e-wallet.
3. Allow users to purchase an e-ticket or apply for a toll pass.
4. Generate a QR code for the pass or ticket.
5. Provide an admin section to manage toll details and approve passes.

**Program:**

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>EasyToll - Toll Payment System</title>
<script
src="https://cdnjs.cloudflare.com/ajax/libs/qrcodejs/1.0.0/qrcode.min.js"></script>
<style>
body{font-family:Arial;margin:0;background:#eef3f8;color:#203047}
header{background:#123c69;color:white;text-align:center;padding:20px}
main{max-width:850px;margin:20px auto;padding:12px}
.card{background:white;padding:18px;margin:15px 0;border-radius:12px;box-shadow:0
3px 10px #ccd3dd}
input,select,button{padding:10px;margin:5px;border:1px solid #bbb;border-radius:5px}
button{background:#1769aa;color:white;border:0;cursor:pointer}
```

```
button:hover{background:#104c7b}
.balance{font-size:24px;color:#087443;font-weight:bold}
#qrcode{margin:15px}
</style>
</head>
<body>
<header>
<h1>EasyToll</h1>
<p>Pay in advance. Travel without waiting.</p>
</header>
<main>
<div class="card">
<h2>User Registration</h2>
<input id="name" placeholder="Full name">
<input id="email" type="email" placeholder="Email">
<input id="vehicle" placeholder="Vehicle number">
<button onclick="register()">Register</button>
<p id="userMessage"></p>
</div>

<div class="card">
<h2>Toll E-Wallet</h2>
<p>Wallet Balance: <span class="balance" id="balance">₹0</span></p>
<input id="amount" type="number" min="1" placeholder="Amount to add">
<button onclick="addMoney()">Add Money</button>
</div>

<div class="card">
```

```
<h2>Buy Toll E-Ticket</h2>
<select id="route">
<option value="Chennai Central Toll">Chennai Central Toll - ₹100</option>
<option value="City Highway Toll">City Highway Toll - ₹150</option>
<option value="Airport Toll">Airport Toll - ₹80</option>
</select>
<button onclick="buyTicket()">Buy E-Ticket</button>
<button onclick="applyPass()">Apply Toll Pass - ₹500</button>
<p id="ticketMessage"></p>
<div id="qrcode"></div>
</div>

<div class="card">
<h2>Transaction History</h2>
<ul id="history"></ul>
</div>

<div class="card">
<h2>Admin Panel</h2>
<p>Approve pending toll pass applications.</p>
<ul id="applications"></ul>
<input id="tollName" placeholder="New toll location">
<input id="tollFee" type="number" min="1" placeholder="Toll fee (₹)">
<button onclick="createToll()">Add Toll Details</button>
<ul id="tollList"></ul>
</div>

<div class="card">
```

```
<h2>Toll Gate Verification</h2>
<input id="verifyId" placeholder="Enter ticket/pass ID">
<button onclick="verify()">Verify Ticket or Pass</button>
<p id="verifyMessage"></p>
</div>
</main>

<script>
let user=null;
let balance=0;
let records=[];
let passes=[];
let tolls=[
 {name:"Chennai Central Toll",fee:100},
 {name:"City Highway Toll",fee:150},
 {name:"Airport Toll",fee:80}
];

function register(){
 const name=document.getElementById("name").value.trim();
 const email=document.getElementById("email").value.trim();
 const vehicle=document.getElementById("vehicle").value.trim();

 if(!name||!vehicle||!/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email)){
  alert("Enter valid name, email and vehicle number.");
  return;
 }
 user={name,email,vehicle};
```

```
 document.getElementById("userMessage").textContent=
   "Welcome "+name+"! Vehicle: "+vehicle;
}

function addMoney(){
 if(!user){alert("Register first.");return;}
 const amount=Number(document.getElementById("amount").value);
 if(!Number.isFinite(amount)||amount<=0){
   alert("Enter a valid amount.");return;
 }
 balance+=amount;
 document.getElementById("balance").textContent="₹"+balance;
 addHistory("Wallet recharge: ₹"+amount);
}

function makeRecord(type,location,fee){
 const id="ET"+Date.now()+Math.floor(Math.random()*1000);
 const record={id,type,location,fee,vehicle:user.vehicle,status:"Active"};
 records.push(record);
 document.getElementById("ticketMessage").textContent=
   type+" created! ID: "+id+" | "+location+" | ₹"+fee;
 document.getElementById("qrcode").innerHTML="";
 if(window.QRCode){
   new QRCode(document.getElementById("qrcode"),{
     text:id,width:140,height:140
   });
 }else{
   document.getElementById("qrcode").textContent="Ticket ID: "+id;
```

```
 }
 addHistory(type+": "+location+" (₹"+fee+")");
}

function buyTicket(){
 if(!user){alert("Register first.");return;}
 const location=document.getElementById("route").value;
 const toll=tolls.find(t=>t.name===location);
 if(!toll){alert("Toll not found.");return;}
 if(balance<toll.fee){alert("Insufficient balance. Add money first.");return;}
 balance-=toll.fee;
 document.getElementById("balance").textContent="₹"+balance;
 makeRecord("E-Ticket",location,toll.fee);
}

function applyPass(){
 if(!user){alert("Register first.");return;}
 const id="PASS"+Date.now();
 passes.push({id,vehicle:user.vehicle,status:"Pending"});
 document.getElementById("ticketMessage").textContent=
  "Pass application submitted. ID: "+id+" (Awaiting admin approval)";
 showApplications();
}

function showApplications(){
 const list=document.getElementById("applications");
 list.innerHTML="";
 passes.forEach(p=>{
```

```
const li=document.createElement("li");
li.textContent=p.id+" | "+p.vehicle+" | "+p.status+" ";
if(p.status==="Pending"){
 const btn=document.createElement("button");
 btn.textContent="Approve";
 btn.onclick=()=>{
  if(!user || balance<500){
   alert("For this demo, register a user and add at least ₹500 to the wallet.");
   return;
  }
  balance-=500;
  document.getElementById("balance").textContent="₹"+balance;
  p.status="Approved";
  records.push({
   id:p.id,type:"Toll Pass",location:"All listed tolls",
   fee:500,vehicle:p.vehicle,status:"Active"
  });
  addHistory("Toll pass approved: ₹500");
  document.getElementById("ticketMessage").textContent=
   "Toll pass approved! ID: "+p.id;
  document.getElementById("qrcode").innerHTML="";
  if(window.QRCode){
   new QRCode(document.getElementById("qrcode"),p.id);
  }
  showApplications();
 };
 li.appendChild(btn);
}
```

```
  list.appendChild(li);
 });
}

function createToll(){
 const name=document.getElementById("tollName").value.trim();
 const fee=Number(document.getElementById("tollFee").value);
 if(!name||!Number.isFinite(fee)||fee<=0){
  alert("Enter a valid toll name and fee.");return;
 }
 if(tolls.some(t=>t.name.toLowerCase()===name.toLowerCase())){
  alert("Toll already exists.");return;
 }
 tolls.push({name,fee});
 const option=document.createElement("option");
 option.value=name;
 option.textContent=name+" - ₹"+fee;
 document.getElementById("route").appendChild(option);
 showTolls();
}

function showTolls(){
 const list=document.getElementById("tollList");
 list.innerHTML="";
 tolls.forEach(t=>{
  const li=document.createElement("li");
  li.textContent=t.name+" - ₹"+t.fee;
  list.appendChild(li);
```

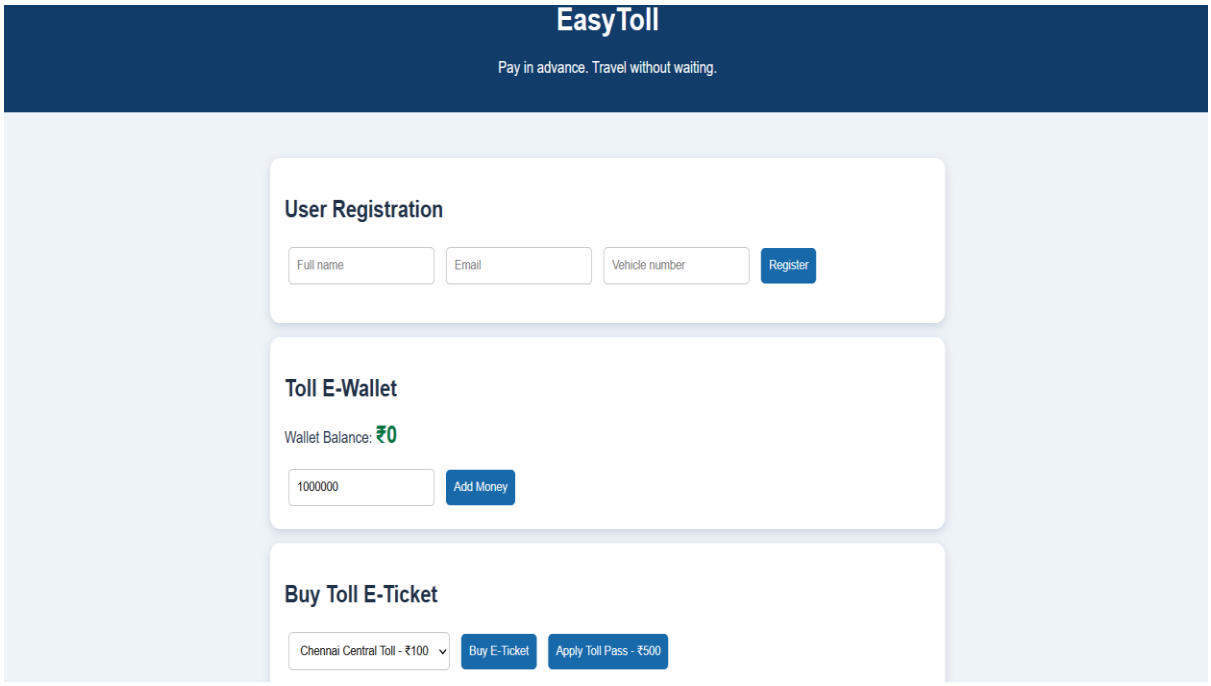
```
 });
}

function addHistory(text){
 const li=document.createElement("li");
 li.textContent=new Date().toLocaleString()+" - "+text;
 document.getElementById("history").prepend(li);
}

function verify(){
 const id=document.getElementById("verifyId").value.trim();
 const record=records.find(r=>r.id===id&&r.status==="Active");
 const pass=passes.find(p=>p.id===id&&p.status==="Approved");
 document.getElementById("verifyMessage").textContent=
  record||pass?"VALID: Ticket/pass approved.":"INVALID: ID not found or not approved.";
}

showTolls();
showApplications();
</script>
</body>
</html>
```

**Output**




**Result:**

The EasyToll Online Toll Payment System was implemented using HTML, CSS, and JavaScript.